Chapter 6 • Trigonometry

Solved Exercise 6.3

1. θ lies in Quadrant I. From each given ratio, find the other five trigonometric ratios.

(i) $\sin\theta=\frac{2}{3}$ Take perpendicular=2 and hypotenuse=3; base=$\sqrt{(3^2-2^2)}=\sqrt{5}$.

Thus $\cos\theta=\frac{\sqrt{5}}{3}$, $\tan\theta=\frac{2}{\sqrt{5}}$, $\operatorname{cosec}\theta=\frac{3}{2}$, $\sec\theta=\frac{3}{\sqrt{5}}$ and $\cot\theta=\frac{\sqrt{5}}{2}$

(ii) $\cos\theta=\frac{3}{4}$ Take base=3 and hypotenuse=4; perpendicular=$\sqrt{(4^2-3^2)}=\sqrt{7}$.

Thus $\sin\theta=\frac{\sqrt{7}}{4}$, $\tan\theta=\frac{\sqrt{7}}{3}$, $\operatorname{cosec}\theta=\frac{4}{\sqrt{7}}$, $\sec\theta=\frac{4}{3}$ and $\cot\theta=\frac{3}{\sqrt{7}}$

(iii) $\tan\theta=\frac{1}{2}$ Take perpendicular=1 and base=2; hypotenuse=$\sqrt{5}$.

Thus $\sin\theta=\frac{1}{\sqrt{5}}$, $\cos\theta=\frac{2}{\sqrt{5}}$, $\operatorname{cosec}\theta=\sqrt{5}$, $\sec\theta=\frac{\sqrt{5}}{2}$ and $\cot\theta=2$.

(iv) $\sec\theta=3$, so $\cos\theta=\frac{1}{3}$ Take hypotenuse=3 and base=1; perpendicular=$\sqrt{8}=2\sqrt{2}$.

Thus $\sin\theta=\frac{2\sqrt{2}}{3}$, $\tan\theta=2\sqrt{2}$, $\operatorname{cosec}\theta=\frac{3}{2\sqrt{2}}$ and $\cot\theta=\frac{1}{2\sqrt{2}}$.

(v) $\cot\theta=\sqrt{(3/2)}=\frac{\sqrt{3}}{\sqrt{2}}$ Take base=$\sqrt{3}$ and perpendicular=$\sqrt{2}$; hypotenuse=$\sqrt{5}$.

Thus $\tan\theta=\sqrt{(\frac{2}{3})}$, $\sin\theta=\sqrt{\frac{2}{5}}$, $\cos\theta=\sqrt{(\frac{3}{5})}$, $\operatorname{cosec}\theta=\sqrt{\frac{5}{2}}$ and $\sec\theta=\sqrt{(\frac{5}{3})}$.

2. Prove $(\sin\theta+\cos\theta)^2=1+2\sin\theta\cos\theta$.

Expand: $\sin^2\theta+2\sin\theta\cos\theta+\cos^2\theta=1+2\sin\theta\cos\theta$.

3. Prove $\frac{\cos\theta}{\sin\theta}=\frac{1}{\tan\theta}$

$\frac{\cos\theta}{\sin\theta}=\cot\theta$

$=\frac{1}{\tan\theta}$

4. Prove $\frac{\sin\theta}{\operatorname{cosec}\theta}\frac{+\cos\theta}{\sec\theta}=1$.

Left side=$\sin^2\theta+\cos^2\theta=1$.

5. Prove $\cos^2\theta-\sin^2\theta=2\cos^2\theta-1$.

$\cos^2\theta-(1-\cos^2\theta)=2\cos^2\theta-1$.

6. Prove $\cos^2\theta-\sin^2\theta=1-2\sin^2\theta$.

$1-\sin^2\theta-\sin^2\theta=1-2\sin^2\theta$.

7. Prove $\frac{1-\sin\theta}{\cos\theta}=\frac{\cos\theta}{1+\sin\theta}$.

$(1-\sin^2\theta)/[\cos\theta(1+\sin\theta)]=\cos^2\theta/[\cos\theta(1+\sin\theta)]$

$=\frac{\cos\theta}{1+\sin\theta}$.

8. Prove $(\sec\theta-\tan\theta)^2=\frac{1-\sin\theta}{1+\sin\theta}$.

$(\sec\theta-\tan\theta)^2=(1-\sin\theta)^2/\cos^2\theta$

$=(1-\sin\theta)^2/[(1-\sin\theta)(1+\sin\theta)]$

$=\frac{1-\sin\theta}{1+\sin\theta}$.

9. Prove $(\tan\theta+\cot\theta)^2=\sec^2\theta\operatorname{cosec}^2\theta$.

$\tan\theta+\cot\theta=\frac{\sin^2\theta+\cos^2\theta}{\sin\theta\cos\theta}=\frac{1}{\sin\theta\cos\theta}$; square both sides.

10. Prove $\frac{\tan\theta+\sec\theta-1}{\tan\theta-\sec\theta+1}=\tan\theta+\sec\theta$.

Use $1=\sec^2\theta-\tan^2\theta=(\sec\theta-\tan\theta)(\sec\theta+\tan\theta)$.

Numerator = $(\tan\theta+\sec\theta)-(\sec\theta-\tan\theta)(\sec\theta+\tan\theta)$

$=(\tan\theta+\sec\theta)(1-\sec\theta+\tan\theta)$.

Cancel the nonzero factor $(\tan\theta-\sec\theta+1)$.

Identity proved.

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11. Prove $\sin^3\theta-\cos^3\theta=(\sin\theta-\cos\theta)(1+\sin\theta\cos\theta)$.
Use $a^3-b^3=(a-b)(a^2+ab+b^2)$, then $\sin^2\theta+\cos^2\theta=1$.

12. Prove $\sin^6\theta-\cos^6\theta=(\sin^2\theta-\cos^2\theta)(1-\sin^2\theta\cos^2\theta)$.
Factor as a difference of cubes in $\sin^2\theta$ and $\cos^2\theta$. The other factor is
$\sin^4\theta+\cos^4\theta+\sin^2\theta\cos^2\theta=(\sin^2\theta+\cos^2\theta)^2-\sin^2\theta\cos^2\theta=1-\sin^2\theta\cos^2\theta$.

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